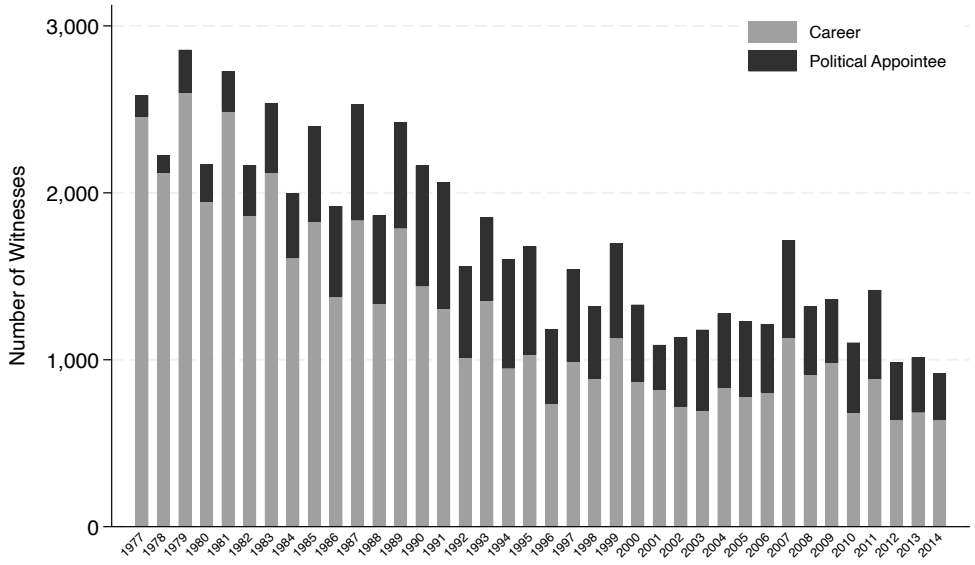
Supplemental Material (not copyedited or formatted) for: Pamela Ban, Ju Yeon Park, Hye Young You. 2026. "Bureaucrats in Congress: The Politics of Inter-branch Information Sharing."
The Journal of Politics 88(2). DOI: https://doi.org/10.1086/735440.

Supporting Information for

Bureaucrats in Congress: The Politics of Interbranch Information Sharing

Appendix A: Additional Figures

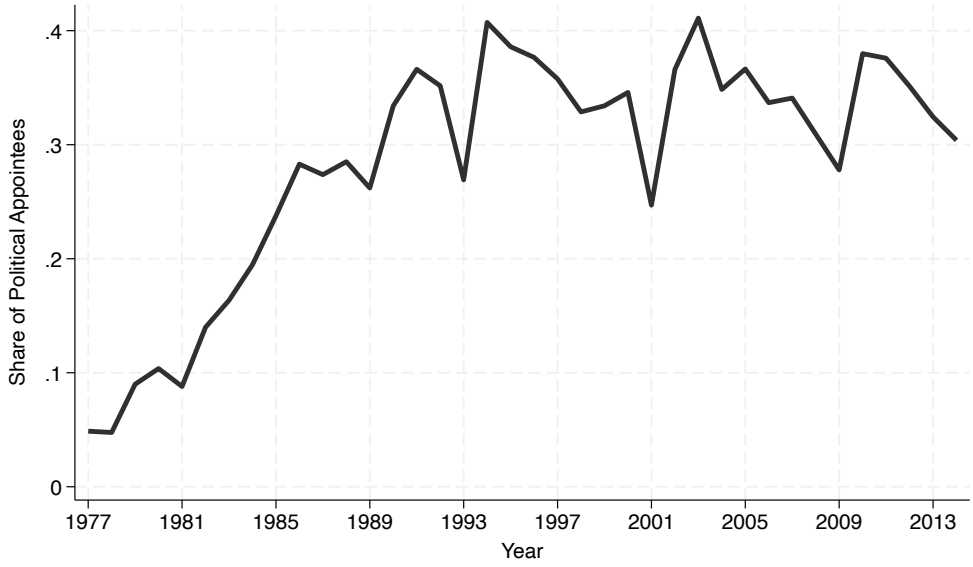
Figure A1: Types of Bureaucratic Witnesses Over Time



Notes: The figure shows the number of career vs. political appointees among bureaucratic witnesses for the period 1977-2014 in the House hearings. We combine the executive departments and other agencies.

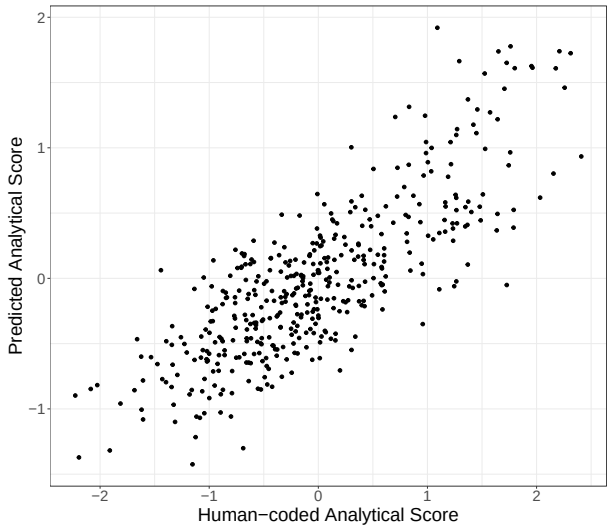
Supplemental Material (not copyedited or formatted) for: Pamela Ban, Ju Yeon Park, Hye Young You. 2026. "Bureaucrats in Congress: The Politics of Inter-branch Information Sharing."
The Journal of Politics 88(2). DOI: https://doi.org/10.1086/735440.

Figure A2: Share of Political Appointees Over Time



Notes: The figure shows the share of political appointees among bureaucratic witnesses for the period 1977-2014 in the House hearings. We combine the executive departments and other agencies.

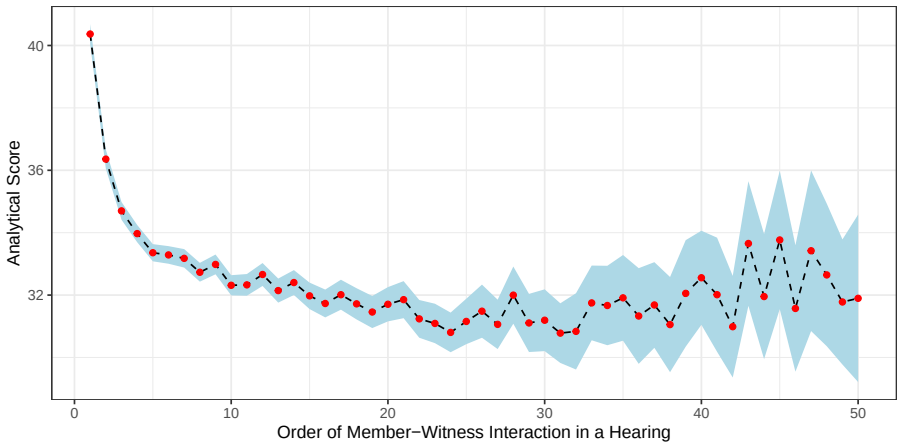
Figure A3: Validation of the Final Ensemble Model



Notes: The graph shows how our final model predicts the human-coded labels of the validation set that was set aside. The Pearson correlation coefficient between the human-coded labels and the model predictions is 0.81, and the Root Mean Squared Errors (RMSE) is 0.53.

Supplemental Material (not copyedited or formatted) for: Pamela Ban, Ju Yeon Park, Hye Young You. 2026. "Bureaucrats in Congress: The Politics of Inter-branch Information Sharing."
The Journal of Politics 88(2). DOI: https://doi.org/10.1086/735440.

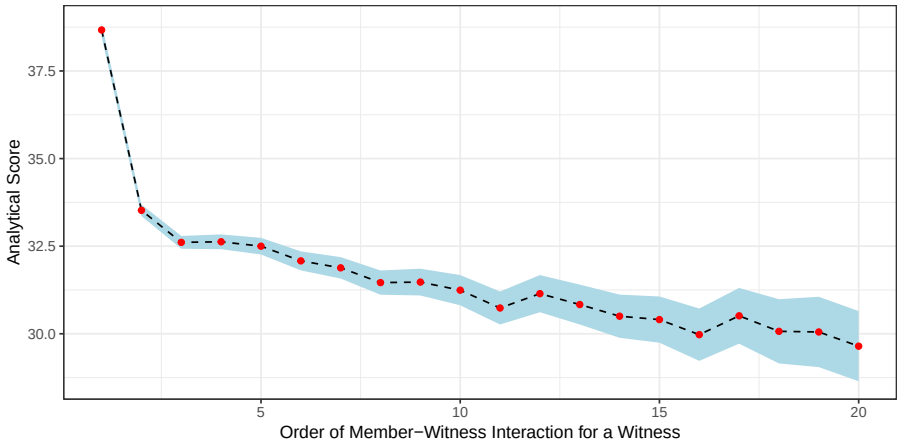
Figure A4: Analytical Information and the Order of Member-Witness Pair



Notes: We counted the order of each member-witness dyad that appeared for the first time in each hearing. The horizontal axis shows the order of these pairs in our data. The vertical axis is the average analytical score of witness testimonies for the corresponding member-witness pair order. Because there are fewer observations as the order number increases, the confidence interval widens significantly, making the computed average scores less reliable. Therefore, we censored the horizontal axis at 50. The graph shows that there is greater transmission of analytical information at an earlier stage in each hearing.

Supplemental Material (not copyedited or formatted) for: Pamela Ban, Ju Yeon Park, Hye Young You. 2026. "Bureaucrats in Congress: The Politics of Inter-branch Information Sharing."
The Journal of Politics 88(2). DOI: https://doi.org/10.1086/735440.

Figure A5: Analytical Information and the Order of Witness Response



Notes: We counted the order of dyadic member-witness interactions for each witness in a given hearing. For example, if a witness interacted with three legislators, we order the three pairs based on which interaction occurred earlier in a hearing. The horizontal axis shows the order of these pairs in our data. The vertical axis is the average analytical score of witness testimonies for the corresponding member-witness pair order. Because there are fewer observations as the order number increases, the confidence interval widens significantly, making the computed average scores less reliable. Therefore, we censored the horizontal axis at 20. The graph suggests that witnesses tend to provide more analytical information in response to questions posed by legislators with whom they interacted earlier in the given hearing session.

Supplemental Material (not copyedited or formatted) for: Pamela Ban, Ju Yeon Park, Hye Young You. 2026. "Bureaucrats in Congress: The Politics of Inter-branch Information Sharing."
The Journal of Politics 88(2). DOI: https://doi.org/10.1086/735440.

Appendix B: Additional Tables

Table B1: Summary Statistics of the Members' Questioning Styles

	Divided		**Unified**	
Variable	Co-Partisan	Out-Partisan	Co-Partisan	Out-Partisan
Grandstanding Score	44.81	42.92	43.47	45.48
Positive Tone Share	0.28	0.24	0.29	0.24
Neutral Tone Share	0.56	0.58	0.57	0.58
Negative Tone Share	0.16	0.18	0.14	0.18
Analytical Word Count	5.86	4.81	5.35	6.05
Word Count	131.52	109.64	123.63	133.06
N	14,716	23,088	14,115	9,883

Notes: The numbers indicate the mean value for each variable. Unit of observation is at member-bureaucrat-hearing level.

Supplemental Material (not copyedited or formatted) for: Pamela Ban, Ju Yeon Park, Hye Young You. 2026. "Bureaucrats in Congress: The Politics of Inter-branch Information Sharing."
The Journal of Politics 88(2). DOI: https://doi.org/10.1086/735440.

Table B2: Summary Statistics of the Variables at the Pair Level

Variables	N	Mean	SD	Min	Max
Panel A: Bureaucratic Witness Analysis					
Analytical Score	61802	33.48	9.33	12.96	64.61
Number of Word (Witness)	61802	152.01	155.21	11.00	2603.00
Political Appointee	61802	0.35	0.48	0.00	1.00
President Out-Partisan	61802	0.53	0.50	0.00	1.00
Majority	61802	0.60	0.49	0.00	1.00
Committee Chair	61802	0.16	0.36	0.00	1.00
Ranking Member	61802	0.06	0.24	0.00	1.00
Ideological Extremism (Member)	61802	0.16	0.13	0.00	1.00
LES	61802	1.20	1.63	0.00	18.69
Dyad Order	61802	11.49	10.90	1	138
Witness Response Order	61802	5.11	4.87	1	52
Grandstanding Score	61802	43.91	11.99	6.12	93.11
Share of Positive Tone	61802	0.26	0.28	0.00	0.99
Share of Neutral Tone	61802	0.57	0.25	0.00	0.98
Share of Negative Tone	61802	0.17	0.18	0.00	0.98
Analytical Word Count (Member)	61802	5.38	5.81	0.00	106.00
Number of Word (Member)	61802	121.79	120.98	10.00	1679.00
Subcommittee	61802	0.63	0.48	0.00	1.00
Referral Hearing	61802	0.10	0.29	0.00	1.00
Number of Witness	61802	5.64	4.49	1.00	76.00
Panel B: Non-Bureaucratic Witness Analysis					
Analytical Score	87261	35.44	11.03	11.80	67.59
Number of Word (Witness)	87261	212.15	238.15	10.00	3564.00
President Out-Partisan	87261	0.53	0.50	0.00	1.00
Majority	87261	0.65	0.48	0.00	1.00
Committee Chair	87261	0.26	0.44	0.00	1.00
Ranking Member	87261	0.06	0.24	0.00	1.00
Ideological Extremism	87261	0.17	0.13	0.00	1.00
LES	87261	1.46	1.90	0.00	18.69
Dyad Order	87261	13.77	11.83	1	151
Witness Response Order	87261	2.74	2.33	1	36
Grandstanding Score	87261	42.30	13.50	3.05	94.08
Share of Positive Tone	87261	0.30	0.32	0.00	0.99
Share of Neutral Tone	87261	0.56	0.29	0.00	0.98
Share of Negative Tone	87261	0.14	0.19	0.00	0.98
Analytical Word Count (Member)	87261	4.48	6.06	0.00	317.00
Number of Word (Member)	87261	106.85	124.58	8.00	4817.00
Subcommittee	87261	0.71	0.45	0.00	1.00
Referral Hearing	87261	0.19	0.39	0.00	1.00
Number of Witness	87261	7.99	9.43	1.00	127.00

Notes: Unit of observation is bureaucrat-politician pair in a given hearing conditional on at least one interaction. The data covers the hearings in the House of Representatives for 1997-2014.

Supplemental Material (not copyedited or formatted) for: Pamela Ban, Ju Yeon Park, Hye Young You. 2026. "Bureaucrats in Congress: The Politics of Inter-branch Information Sharing."
The Journal of Politics 88(2). DOI: https://doi.org/10.1086/735440.

Table B3: Bureaucratic Witnesses' Provision of Analytical Information: Including All Political Appointees

	Legislative			**Oversight**		
	(1)	(2)	(3)	(4)	(5)	(6)
President Out-Partisan	-1.266***	-1.118***	-1.084***	-1.432***	-1.184***	-1.085***
	(0.150)	(0.140)	(0.165)	(0.167)	(0.153)	(0.172)
Political Appointee	0.123	-0.271		0.640**	0.0996	
	(0.199)	(0.187)		(0.216)	(0.197)	
President Out-Partisan X	-0.658**	-0.382	-0.305	-1.061***	-0.663**	-0.729**
Political Appointee	(0.223)	(0.206)	(0.244)	(0.232)	(0.215)	(0.238)
Committee, Issue, President FEs	✓	✓	✓	✓	✓	✓
Agency FE	✓	✓		✓	✓	
Witness-Hearing FE			✓			✓
Member Question Controls		✓	✓		✓	✓
Mean Outcome Measure	33.3	33.3	33.3	33.5	33.5	33.5
N	30117	30117	30117	27401	27401	27401
adj. R^2	0.098	0.190	0.325	0.112	0.203	0.333

Notes: $*$ p<0.05, $**$ p<0.01, $***$ p<0.001. Standard errors are clustered at the hearing-level. Member- and hearing-level control variables are included in the regression but not presented in the table. The *Political Appointee* variable includes all types of political appointees: PAS, PA, and C.

Table B4: Controlling for the Order of Witness Response

	Legislative			**Oversight**		
	(1)	(2)	(3)	(4)	(5)	(6)
President Out-Partisan	-1.345***	-1.185***	-1.183***	-1.438***	-1.183***	-1.151***
	(0.141)	(0.131)	(0.155)	(0.162)	(0.149)	(0.167)
Political Appointee	0.448*	-0.0934		0.775***	0.155	
	(0.213)	(0.198)		(0.218)	(0.196)	
President Out-Partisan X	-0.369	-0.111	-0.0759	-0.876***	-0.413	-0.625*
Political Appointee	(0.229)	(0.211)	(0.254)	(0.240)	(0.221)	(0.247)
Witness Response Order	-0.374***	-0.356***	-0.336***	-0.254***	-0.276***	-0.265***
	(0.0282)	(0.0282)	(0.0285)	(0.0171)	(0.0158)	(0.0177)
Committee, Issue, President FEs	✓	✓	✓	✓	✓	✓
Agency FE	✓	✓		✓	✓	
Witness-Hearing FE			✓			✓
Member Question Controls		✓	✓		✓	✓
(ln) Mean Outcome Measure	33.3	33.3	33.3	33.5	33.5	33.5
N	30117	30117	30117	27401	27401	27401
adj. R^2	0.103	0.199	0.320	0.119	0.214	0.331

Notes: $*$ p<0.05, $**$ p<0.01, $***$ p<0.001. Standard errors are clustered at the hearing-level. Member- and hearing-level control variables are included in the regression but not presented in the table.

Supplemental Material (not copyedited or formatted) for: Pamela Ban, Ju Yeon Park, Hye Young You. 2026. "Bureaucrats in Congress: The Politics of Inter-branch Information Sharing."
The Journal of Politics 88(2). DOI: https://doi.org/10.1086/735440.

Table B5: Bureaucrat's Provision of Analytical Information: Divided vs. Unified Government

	Divided	**Unified**
	(1)	(2)
President Out-Partisan	-0.867***	-1.085***
	(0.198)	(0.297)
President Out-Partisan X	-0.855**	-0.276
Political Appointee	(0.292)	(0.444)
Controls	✓	✓
Witness-Hearing FE	✓	✓
Mean Outcome Measure	33.1	34.5
N	19108	9041
adj. R^2	0.352	0.285

Notes: The sample is oversight hearings in the House. $*$ p<0.05, $**$ p<0.01, $***$ p<0.001. Standard errors are clustered at the hearing-level. Member- and hearing-level control variables are included in the regression but not presented in the table.

Table B6: Bureaucrats from the Executive Departments vs. Other

	Executive Department		**Other**	
	(1) Legislative	(2) Oversight	(3) Legislative	(4) Oversight
President Out-Partisan	-1.156***	-1.113***	-1.298***	-1.336***
	(0.178)	(0.186)	(0.301)	(0.309)
President Out-Partisan X	-0.128	-0.682*	0.0200	-0.536
Political Appointee	(0.297)	(0.288)	(0.483)	(0.465)
Controls	✓	✓	✓	✓
Witness-Hearing FE	✓	✓	✓	✓
Mean Outcome Measure	33.4	33.7	33.1	33.2
N	23045	21201	7072	6200
adj. R^2	0.326	0.327	0.321	0.355

Notes: $*$ p<0.05, $**$ p<0.01, $***$ p<0.001. Standard errors are clustered at the hearing-level. Member- and hearing-level control variables are included in the regression but not presented in the table.

Supplemental Material (not copyedited or formatted) for: Pamela Ban, Ju Yeon Park, Hye Young You. 2026. "Bureaucrats in Congress: The Politics of Inter-branch Information Sharing."
The Journal of Politics 88(2). DOI: https://doi.org/10.1086/735440.

Table B7: Bureaucrats from the Executive Departments vs. Other - Treating the EPA as the Executive Department

	Executive Department		**Other**	
	(1) Legislative	(2) Oversight	(3) Legislative	(4) Oversight
President Out-Partisan	-1.175***	-1.096***	-1.221***	-1.420***
	(0.174)	(0.183)	(0.323)	(0.320)
President Out-Partisan X	-0.132	-0.739**	0.0260	-0.211
Political Appointee	(0.287)	(0.282)	(0.534)	(0.493)
Controls	✓	✓	✓	✓
Witness-Hearing FE	✓	✓	✓	✓
Mean Outcome Measure	33.3	33.6	33.3	33.4
N	23983	21949	6134	5452
adj. R^2	0.330	0.331	0.307	0.345

Notes: $*$ p<0.05, $**$ p<0.01, $***$ p<0.001. Standard errors are clustered at the hearing-level. Member- and hearing-level control variables are included in the regression but not presented in the table.

Table B8: Incorporating Agency Ideology: President Aligned vs. Not-Aligned Agencies

	Legislative	**Oversight**
	(1)	(2)
President Out-Partisan	-0.969***	-1.187***
	(0.152)	(0.166)
Political Appointee	-0.522**	-0.368*
	(0.164)	(0.158)
President-Aligned	0.298	0.00688
	(0.178)	(0.199)
President Out-Partisan X	-0.612**	-0.582**
President -Aligned	(0.207)	(0.222)
Controls	✓	✓
Agency, Issue, Committee, President FEs	✓	✓
Mean Outcome Measure	33.4	33.6
N	30117	27401
adj. R^2	0.190	0.203

Notes: $*$ p<0.05, $**$ p<0.01, $***$ p<0.001. Standard errors are clustered at the hearing-level. In all regressions, issue-, committee-, and president-fixed effects and other control variables are included.

Supplemental Material (not copyedited or formatted) for: Pamela Ban, Ju Yeon Park, Hye Young You. 2026. "Bureaucrats in Congress: The Politics of Inter-branch Information Sharing."
The Journal of Politics 88(2). DOI: https://doi.org/10.1086/735440.

Table B9: Non-Bureaucratic Witnesses' Provision of Analytical Information

	Legislative			**Oversight**		
	(1)	(2)	(3)	(4)	(5)	(6)
President Out-Partisan	-0.169	-0.0362	0.0582	-0.176	-0.158	-0.100
	(0.106)	(0.146)	(0.218)	(0.159)	(0.216)	(0.310)
Majority	0.727***	0.746***	0.767**	0.582***	0.468*	0.618
	(0.118)	(0.164)	(0.251)	(0.165)	(0.222)	(0.332)
Committee Chair	3.772***	3.174***	3.139***	4.351***	3.960***	3.486***
	(0.152)	(0.220)	(0.328)	(0.216)	(0.298)	(0.434)
Ranking Member	-0.0472	-0.675*	-0.958*	0.541*	0.110	0.123
	(0.214)	(0.299)	(0.466)	(0.263)	(0.376)	(0.572)
Ideological Exstremism	-1.053**	-1.058	-0.280	-1.251*	-1.021	-1.540
	(0.403)	(0.564)	(0.869)	(0.539)	(0.742)	(1.177)
LES	0.0526	-0.0435	-0.117	0.102*	-0.0130	-0.0399
	(0.0371)	(0.0527)	(0.0830)	(0.0491)	(0.0646)	(0.0977)
Subcommittee	0.323*			0.922***		
	(0.163)			(0.210)		
Referral Hearing	-0.0486			0.305		
	(0.147)			(0.265)		
Number of Witness	0.108***			0.0759***		
	(0.0181)			(0.0124)		
Dyad Order	-0.106***	-0.302***	-0.342***	-0.0917***	-0.271***	-0.296***
	(0.00864)	(0.0171)	(0.0310)	(0.0126)	(0.0200)	(0.0373)
Grandstanding Score	-0.156***	-0.116***	-0.134***	-0.165***	-0.128***	-0.148***
	(0.00617)	(0.00865)	(0.0131)	(0.00789)	(0.0103)	(0.0156)
Share of Positive Tone	8.976***	7.258***	7.475***	8.603***	7.188***	7.469***
	(0.205)	(0.290)	(0.448)	(0.274)	(0.383)	(0.591)
Share of Negative Tone	-0.542*	-0.502	-0.921	-0.207	0.0325	0.325
	(0.277)	(0.389)	(0.630)	(0.366)	(0.519)	(0.817)
Analytical Word Count	0.0468*	0.0734*	0.00303	0.0999***	0.132***	0.0939
	(0.0218)	(0.0306)	(0.0464)	(0.0235)	(0.0320)	(0.0526)
(ln) Number of Word	0.0138***	0.0115***	0.0153***	0.0132***	0.0111***	0.0130***
	(0.00101)	(0.00140)	(0.00216)	(0.00130)	(0.00180)	(0.00296)
Witness Type, Issue, Committee, President FEs	✓			✓		
Witness-Hearing FE		✓	✓		✓	✓
Bureaucrat in Hearing			✓			✓
Mean Outcome Measure	35.3	35.3	35.9	35.5	35.5	35.9
N	52217	52217	27383	28168	28168	14241
adj. R^2	0.222	0.350	0.352	0.232	0.359	0.358

Notes: $*$ p<0.05, $**$ p<0.01, $***$ p<0.001. Standard errors are clustered at the hearing-level.

Supplemental Material (not copyedited or formatted) for: Pamela Ban, Ju Yeon Park, Hye Young You. 2026. "Bureaucrats in Congress: The Politics of Inter-branch Information Sharing."
The Journal of Politics 88(2). DOI: https://doi.org/10.1086/735440.

Table B10: Number of Question-Answer Pairs at the Aggregated-Level

	(1)	(2)	(3)	(4)	(5)	(6)
	Bureaucrat			**Non-Bureaucrat**		
	Total	Co-Partisan	Out-Partisan	Total	Co-Partisan	Out-Paritsan
Divided Government	1.003*	-1.007***	2.010***	1.810***	-1.967***	3.777***
	(0.400)	(0.225)	(0.301)	(0.166)	(0.253)	(0.348)
Issue, Committee, President FEs	✓	✓	✓	✓	✓	✓
Mean Outcome Measure	7.3	3.3	3.9	9.4	4.3	5.1
N	10753	10753	10753	10753	10753	10753
adj. R^2	0.275	0.260	0.271	0.349	0.296	0.358

Notes: $*$ p<0.05, $**$ p<0.01, $***$ p<0.001. The unit of observation is hearing. Standard errors are clustered at the committee-level. Control variables (oversight hearing, subcommittee, referral hearing, number of witness) are included.

Table B11: Total Amount of Analytical Information at the Aggregated-Level

	(1)	(2)	(3)	(4)	(5)	(6)
	Bureaucrat			**Non-Bureaucrat**		
	Total	Co-Partisan	Out-Partisan	Total	Co-Partisan	Out-Paritsan
Divided Government	-51.82**	-164.3***	112.5***	47.40	-495.9***	543.3***
	(17.90)	(19.48)	(18.56)	(23.49)	(62.96)	(61.35)
Issue, Committee, President FEs	✓	✓	✓	✓	✓	✓
Mean Outcome Measure	492.0	250.5	241.4	958.3	455.4	502.8
N	10753	10753	10753	10753	10753	10753
adj. R^2	0.220	0.197	0.193	0.452	0.314	0.405

Notes: The outcome is the total analytical information weighted by the number of words in the testimony at the hearing level. $*$ p<0.05, $**$ p<0.01, $***$ p<0.001. The unit of observation is hearing. Standard errors are clustered at the committee-level. Control variables (oversight hearing, subcommittee, referral hearing, number of witness) are included.

Supplemental Material (not copyedited or formatted) for: Pamela Ban, Ju Yeon Park, Hye Young You. 2026. "Bureaucrats in Congress: The Politics of Inter-branch Information Sharing."
The Journal of Politics 88(2). DOI: https://doi.org/10.1086/735440.

Appendix C: Heterogeneous Effects

We examine whether the effect of partisan alignment on bureaucrats' provision of analytical information varies by issue polarization and types of committees. First, for issue polarization, we create a vote-based measure of issue polarization for 20 major issue areas, defined by the Comparative Agenda Project (Baumgartner and Jones 2015). Within each issue area, for each passage or amendment roll-call vote, we calculate the percentage of each party voting yea for these roll call votes and then calculate the absolute difference between these percentages. We take the mean of these absolute differences across all roll-call votes in that issue area in a given Congress to generate the issue polarization score by issue and Congress.

Then, for each Congress, we divide the issues into three groups (Bottom, Middle, and Top) based on the degree of polarization between Democrats and Republicans. We merge the Congress-specific issue polarization score to our hearing data based on the major issue code for each hearing. Then, we run separate regressions of equation (3) with witness-hearing FEs for three different groups based on the issue polarization categories. Figure C1 shows that the regression coefficients of the variable *President Out-Partisan* and bureaucrats provide significantly less analytical information when they appear in hearings that address highly polarized issues and this pattern is more salient in legislative hearings.

We also examine whether the effect of the partisan alignment varies by the types of committees. Deering and Smith (1997) divide committees into three types: (1) prestigious (Appropriations, Budget, Rules, and Ways and Means); (2) policy (Financial Services, Education and Labor, Energy and Commerce, Foreign Affairs, Judiciary, Government Oversight); and (3) constituency service (Agriculture, Armed Services, Natural Resources, Science, Small Business, and Veterans Affairs). Figure C2 shows the coefficients of the *President Out-Partisan* variable on bureaucrat's provision of analytical information across different types of committees. The negative effect of partisan misalignment is larger in prestigious and policy committees than constituent service committees and this pattern is more salient in legislative hearings.

Supplemental Material (not copyedited or formatted) for: Pamela Ban, Ju Yeon Park, Hye Young You. 2026. "Bureaucrats in Congress: The Politics of Inter-branch Information Sharing."
The Journal of Politics 88(2). DOI: https://doi.org/10.1086/735440.

Figure C1: Heterogeneous Effects of "President Out-Partisan" by Issue Polarization

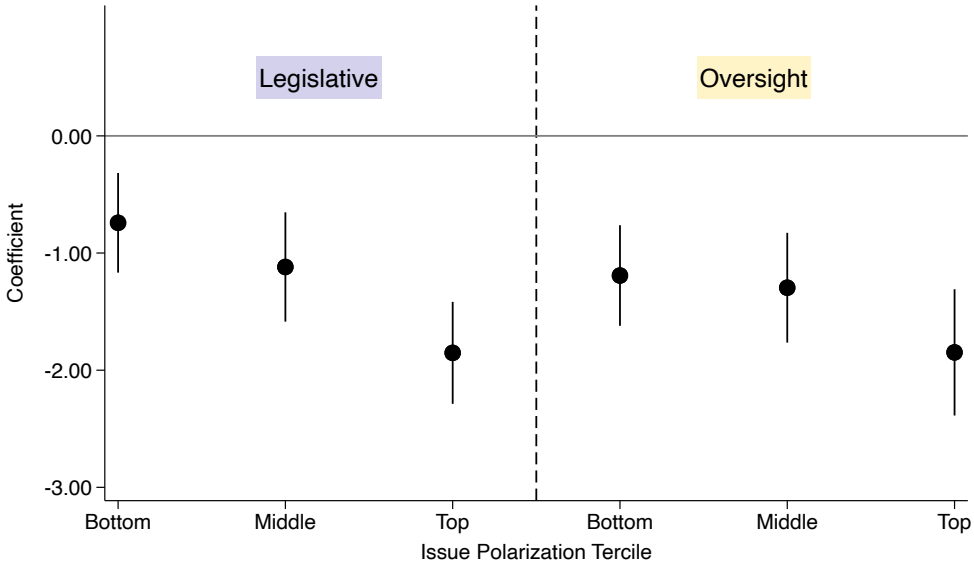
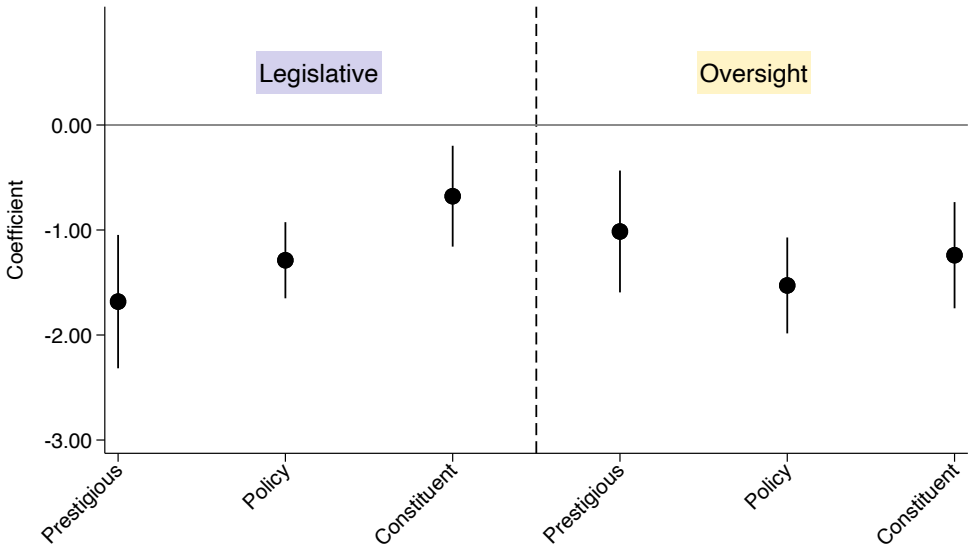


Figure C2: Heterogeneous Effects of "President Out-Partisan" by Committee Type



Supplemental Material (not copyedited or formatted) for: Pamela Ban, Ju Yeon Park, Hye Young You. 2026. "Bureaucrats in Congress: The Politics of Inter-branch Information Sharing."
The Journal of Politics 88(2). DOI: https://doi.org/10.1086/735440.

Appendix D: Measuring Analytical Information

Coding Instructions for MTurk Workers

This task involves reading two statement excerpts made by witnesses invited to testify at congressional committee hearings held from 1997 to 2018. Researchers will use your responses to better understand the "tone" of each statement.

Your job is to read both statements and select the one that is relatively more analytical. To give you some background knowledge, congressional committees hold hearings to collect policy-relevant information from external experts, bureaucrats, citizens or any groups that can be affected by policies that committees consider (e.g. trade associations, farmers, veterans, etc.). In a typical hearing, witnesses give an opening statement and then answer questions that committee members ask during the Q and A session. Witness testimonies transmit various types of information to committee members (e.g. analytical information, political information on preferences of certain group of people, etc.). Our study specifically focuses on identifying and measuring analytical information that witnesses provide. To do so, you will help us by comparing two randomly selected excerpts from witness testimonies and choosing the one that sounds relatively more analytical. We define a testimony as analytical if it contains statements that are fact-based, objective or research/data driven. In contrast, a non-analytical testimony tends to contain statements that are based on personal experience or opinion (which scholars call "ordinary knowledge" as opposed to "expert knowledge"), subjective, or normative. **Your performance will be monitored as you complete these HITs. We will reject all work done by workers who provide poor quality answers.** Do not allow your own political opinion to influence your decision. Your goal is to select the statement that other workers would also recognize as more analytical.

A statement is analytical if it is . . .
1) Fact-based
2) Verifiable (Based on research or data driven analysis)
3) Objective

(Example) Fact-based statement:
"I have over 27 years of experience in the field of community and economic development. The authority I represent has approximately 1,300 public housing units. We administer 4,600 housing choice vouchers. We manage market-rate units and two office buildings. In 2010, we opened our housing choice voucher waiting list for only 5 days and received over 6,000 applications. Our public housing waiting lists are currently at 130 percent of our total units."

(Example) Verifiable (Based on research or data driven analysis):
"The available evidence indicates that the response of individuals to increasing amounts of THC is much more variable than it is for alcohol, so with alcohol, we have a considerable body of evidence that can place risk odds at increasing levels of blood alcohol content. For example, .08 blood alcohol content is associated with about four times the crash risk of a sober person.

The average arrest is .15 THC. That's associated with about 15 times the crash risk. Beyond a– some broad confirmation that higher levels of THC are generally associated with higher levels of impairment, a more precise association of various THC levels and degrees of impairment are not yet available."

"There are several options in some of the background in the literature, everything from taking a budget level and determining what different levels of performance you can get for that same budget amount versus different levels of performance for different budget level amounts versus cost agency or even intraagency tradeoffs among requirements and budget responsibilities. So, what we need to do from a piloting standpoint is look at these and say how can we test those theories in various ways."

(Example) Objective:

"When projects are authorized, when there is a Chief's Report and the Congress authorizes a project, the economic analysis that is done on that calculates a benefit to cost ratio. And that benefit to cost ratio is based on a 3.125 discount rate."

A statement is non-analytical if it is . . .

1) Opinion-based/Normative
2) Anecdotal or experiential (Based on personal experience)
3) Subjective or preferential (Revealing preferences of certain groups)
4) Procedural statements
5) All the statements that do not contain analytical information as defined earlier

(Example) Opinion-based/normative:

"We should do it this year. But we should adjust the system so that we get ready for 2017 when more money is going out than coming in, and we can do it."

(Example) Anecdotal or experiential (Based on personal experience):

"When Michael came home that night and I confronted him and was talking to him, he had eye contact like we do now. But when he was sitting on the sofa and nobody was confronting him, he was comatose. He was in the ozone. He was sitting with his mouth hanging open, staring at the door. I knew that there was something wrong with him that night. I could tell that he had taken something."

"I guess we mistakenly believed that it was a secret location, and the only people who knew about it were the EOD staff from both SFPD, the FBI and the Sheriff's Office. Unbeknownst to us, this particular individual, and I won't say too much, but was a plumber in that area and apparently had seen the officers going into that area and perhaps followed them in."

(Example) Subjective or preferential (Revealing preferences of certain groups):

"–that we try to organize that under FEHB because there has been a concern from the employees of not moving away from FEHB. From our perspective, we're okay to wait, as long as we get the savings. The savings are what's key to us. If I could put a chart up here."

Supplemental Material (not copyedited or formatted) for: Pamela Ban, Ju Yeon Park, Hye Young You. 2026. "Bureaucrats in Congress: The Politics of Inter-branch Information Sharing."
The Journal of Politics 88(2). DOI: https://doi.org/10.1086/735440.

(Example) Procedural statements:

"Thank you very much, Mr. Souder, and your staff for helping to deal me in today. I found out about this yesterday morning, and I'm pleased to be here. I am a former college administrator and teacher. My name is Dean, but I was one once."

In summary, consider that all statements can be placed on a continuum ranging from 0 to 100, where 0 is the most extreme non-analytical statement and 100 is the most extreme analytical statement. Some statements can be a mixture of analytical and non-analytical statements; some may be moderately analytical. Consider that these instances can be placed between the two extreme ends of the continuum.

For each HIT, you will receive two speech extracts. Your task is to read both and select which of the two statements is more analytical in the following manner:

If statement A is...	If statement B is...	Then, choose
Analytical	Non-analytical	Statement A
Analytical	Analytical	The one that is more explicitly factual/verifiable/objective
Non-analytical	Non-analytical	The one that is more explicitly opinion-based/experiential/subjective

Please read each statement carefully and judge each by the standards listed above and the information in the text. DO NOT make your judgments on your own knowledge of a person or a policy in question or on definitions of analytical and non-analytical statements different from those listed above. Your performance will be monitored as you complete these HITs. We will reject all work done by workers who provide poor quality answers.

This training module has two parts.

In Part 1, we will provide **5 practice HITs** followed by instructions about how the statements need to be coded.

In Part 2, we will give you **5 test HITs** to complete. To receive qualification for the **Compare Witness Testimony task 2022**, you must complete **4 out of 5 of these test HITs correctly**.

Labeling Process

The sampling of the training paragraphs was a two-stage process. First, we originally planned sampling 3,300 paragraphs from the corpus of witness testimonies. To do this, we randomly selected 3,300 hearings and took only the witness testimonies. To facilitate online workers' comparison of paired paragraphs, the length of paragraphs to constitute the training set was controlled through the following process: 1) For the statements containing multiple paragraphs and more than 150 words, we divided each statement into paragraphs, but we skipped the paragraphs containing less than 50 words to keep them together with the following paragraph so it was long enough; 2) then, the paragraphs containing less than 50 words or more than 150 words were removed. From the remaining paragraphs, we randomly selected 3,300 paragraphs. Each paragraph appeared 20 times in the pair-wise comparisons generating 33,000 comparison tasks or HITs. Using the online workers' binary choices on these tasks and the `labelR` software (Carlson and Montgomery, 2017), we fit a Bradley-Terry model to generate a continuous, human-coded score or label for the 3,300

Supplemental Material (not copyedited or formatted) for: Pamela Ban, Ju Yeon Park, Hye Young You. 2026. "Bureaucrats in Congress: The Politics of Inter-branch Information Sharing."
The Journal of Politics 88(2). DOI: https://doi.org/10.1086/735440.

paragraphs.

However, our machine learning models fit on the random sample of the 3,000 paragraphs could not predict the rest of the 300 paragraphs well. We suspected that this is because the training set did not contain enough variation in the analytical information. Indeed, when we measure this concept using the dictionary of analytical information used in Ban, Park, and You (2023), the measurement has a highly skewed distribution with few statements scoring high. To solve this issue, we decided to repeat the labeling procedure by over-sampling the statements containing the words in their dictionary.

In the second stage, we used random-block sampling to select 1,000 paragraphs to be labeled. For the pre-processed paragraphs from the 3,300 hearings, we computed the proportion of analytical words in the dictionary. Then, we partitioned the paragraphs into four blocks based on this preliminary measurement with three cut-points: 0.05, 0.1 and 0.15. The number of blocks and the cut-points were selected to ensure that we had a high enough number of statements in the block with the highest proportion of analytical words when the equal number of statements are selected for each block. Then, we randomly selected 250 statements for each of the four blocks.

In doing so, we included statements that were labeled in the first stage so they could be used as a bridge to help the Bradley-Terry model learn the relative strength of the analytical information for the paragraphs that were labeled only in the first stage and those labeled only in the second stage. As the paragraphs labeled in the first stage were mostly populated in the blocks featuring low proportions of analytical words, we sampled 150 paragraphs of them for each of the first and second blocks. This renders 100 new paragraphs to be randomly selected for these two blocks. For the third block, we included all 66 paragraphs from those labeled in the first stage and selected 184 new paragraphs. For the fourth block with the proportion of analytical words to be greater than or equal to 0.15, all five paragraphs from those labeled in the first stage and 245 new paragraphs were included. In summary, 371 paragraphs from the first labeling process and 629 new paragraphs were labeled in the second phase.

Then, we fit a Bradley-Terry model on the combination of all the 43,000 HITs collected from the first and second phases to generate our human-coded score of the analytical information for the 3,929 paragraphs.

Learning and Predicting the Analytical Information

First, we pre-processed the corpus by lowercasing, removing stop words, and stemming. However, we decided to keep numbers as they can be an important feature of analytical information. Also, we included both unigrams and bigrams as we confirmed that the prediction performance of the model improved by including bigrams in addition to unigrams. For this test, Kernlab's support vector machine was used as it quickly fits and has relatively high prediction performance.

Second, we constructed document-level matrices using two approaches: term-document frequency (TDF) and doc2vec. For the TDF matrix, we included only the most frequent 2,000 terms due to the large size of our corpus.

Supplemental Material (not copyedited or formatted) for: Pamela Ban, Ju Yeon Park, Hye Young You. 2026. "Bureaucrats in Congress: The Politics of Inter-branch Information Sharing."
The Journal of Politics 88(2). DOI: https://doi.org/10.1086/735440.

Third, we randomly selected 3,500 paragraphs as a training set and held out 429 paragraphs to validate the final model. Using the paragraphs in the training set, we fit the four best performing machine learning models out of the six used in Park (2021) as our data and her data are from the same source: House hearing transcripts. The four models are support vector machine (SVM), Kernlab's support vector machine (KSVM), LASSO, and Gradient Boosting Machine (GBM). These models were fit on each of the two document matrices totaling eight models. The tuning parameters for each of the eight models were chosen through a grid search. For this, Kernlab's support vector machine was used.

Fourth, we used the `EBMAforecast` R package (Montgomery, Hollenbach, and Ward, 2012) to conduct ensemble Bayesian model averaging to reach a final model that basically aggregates all eight models by assigning weights to them to optimize the model prediction. Montgomery, Hollenbach, and Ward (2012) reports that this method achieves better model prediction than any single best machine learning model. Six models received non-zero weights. Table D1 presents the tuning parameters and weights assigned for each of the eight models.

Table D1: Machine Learning Models

Document Matrix	Model	Parameters	Weight
TDF	SVM	cost = 2	0.132
TDF	KSVM	epsilon = 0.1	0.164
TDF	LASSO	nlambda = 200	0.172
TDF	GBM	shrinkage = 0.1	0.061
Doc2vec	SVM	cost = 2	0
Doc2vec	KSVM	epsilon = 0.1	0.291
Doc2vec	LASSO	nlambda = 200	0
Doc2vec	GBM	shrinkage = 0.1	0.181

Using the final model, we predicted the score for the entire corpus and rescaled the score to range from 0 to 100.

Validation of the Measurement

This section validates the measurement statistically and substantively. First, to validate the human-coded labels substantively, below we present the five most analytical and least analytical paragraphs from the human-coded set.

The most analytical paragraphs (in descending order)

[1] "I can probably take that, sir. For MIDRP, there is about $430,000. For the specific on wound infections, there is $895,000. U.S. Navy wound infection research also gets money. I don't have the exact number right here. USUHS has a little over $4 million. For congressional special interest projects on wound infection, there is almost $12 million. SBIR project is about $3.7 million. Dr. Smith spoke about the Defense health programs and then war supplemental intermural projects, there is about another $2.5 million, sir."

Supplemental Material (not copyedited or formatted) for: Pamela Ban, Ju Yeon Park, Hye Young You. 2026. "Bureaucrats in Congress: The Politics of Inter-branch Information Sharing."
The Journal of Politics 88(2). DOI: https://doi.org/10.1086/735440.

[2] "I'm not sure I have those numbers for seven years. I can tell you that during the last two years, that number is in the range of $147 million of State money. That includes a Clean Water Management Trust Fund. We put about 6.5 percent of remaining funds after the budget is complete into a fund and that's anywhere from $40 to $50, $55 million a year. And, in addition to that, we just, of course, passed the Clean Water Responsibility Act. We've significantly increased our ag share program, working with the farmers on BMPs and so forth; so, $147 million if you total that."

[3] "Early data for cyber Monday 2017 by Adobe Analytics indicate that, collectively, shoppers spent almost $3.4 billion on online purchases, a 17 percent increase over last year. Looking at the underlying data, over 50 percent of the virtual store visits and 40 percent of the revenue were made from tablets or smartphones, an increase of 20 percent and 41 percent respectively over last year. This could indicate that the online shopping experience is becoming more frictionless and shoppers are feeling more secure with online transactions."

[4] "The official service cost position for production is $39.8 billion. As I explained, sir, while you were out, we put together what's called an, we used the Air Force cost analysis group to develop the service cost position. The group, in doing their analysis and developing an official cost estimate for this program, estimated the cost at $40.8 billion, which included $1 billion of risk in the out years, of risk that was unidentified. Without that $1 billion for risk, the estimate is $39.8 billion."

[5] "Mr. Taylor, I believe I have an answer to your question. In 2004, the two polar icebreakers cost over $3 million in fuel costs, $3,039,000. In 2005, both the Polar Star and Krasin together cost $1,720,000 for fuel. Breaking that down, the Polar Star which had limited service during that campaign, the fuel cost was $1,057,000, and the cost of the fuel for the Krasin was $662,739."

The least analytical paragraphs (in ascending order)

[1] "OK. I will wrap it up there. With that, I just want to thank you. And I appreciate the opportunity to be here today. This is something I am very passionate about, and I have a lot more I want to share, but a lot of it is in my written testimony. So I appreciate the opportunity, Mr. Chairman, thank you."

[2] "I do have a problem here and I do share my colleagues' concerns with this situation in the VA. And what is the consequence for those staff that are not reporting or are not taking their duty as they should, they are not properly carrying out that responsibility, what is the consequence for them directly?"

[3] "I would say yes and amen to that. There are other things that need to happen in addition to that. I think not just middle class people are concerned about crime. All people are concerned about crime. Poor people are concerned about crime as well. I think the way Jolice Wilson talks about it is—-"

[4] "No, it is not my view of that sort at all. And I would be happy that they would be very well paid. My only perspective is that, in terms of the constitutional purpose, our focus should be on the

Supplemental Material (not copyedited or formatted) for: Pamela Ban, Ju Yeon Park, Hye Young You. 2026. "Bureaucrats in Congress: The Politics of Inter-branch Information Sharing."
The Journal of Politics 88(2). DOI: https://doi.org/10.1086/735440.

production of output. Now, obviously, a well-compensated artist and musician class is probably important for long-run copyright output of creative works—-"

[5] "I hadn't heard that we had that problem before. You mentioned earlier the comment about the dike. I wish I could say more, and I would like to get back to the Committee on that point. Just so that I don't give you an impression that it's all OK or it's all bad, I'd rather go back and talk to my—-"

The paragraphs that received the highest scores tend to present verifiable information frequently referring to numeric figures whereas those scored the lowest points tend to be procedural statements, normative statements presenting their perspectives, or statements expressing uncertainty about a policy situation. Therefore, these two sets of paragraphs provide us with confidence that the human-coding process was conducted by closely capturing the concept that we intended to measure as described in the coding instructions.

Following the suggestions from Park and Montgomery (2024), we took an additional step to validate the human-coded scores by constructing our own coding on a 5-point scale on a random subset of 120 paragraphs that were labeled by online-workers. Then, we compared the crowd-sourced score to our 5-point scale coding to double-check if the online workers coded paragraphs in a way consistent with our conceptualization of analytical information. The Pearson correlation coefficient between the two measurements is 0.906, which provides both statistical and substantive validation of our human-coding process.

Second, we validated the analytical score predicted for the entire corpus. Below we present the ten most and least analytical statements, respectively. As some of the most analytical statements are extremely long, here, we report only those with 150 words or less.

The most analytical statements (in descending order)

[1] "For the joint NBC defense program, which is the program that I manage, in the area of very basic research–this is laboratory-level research for chem-bio–about $33.2 million for fiscal year 2001; in the area of applied research, $73.6 million; for advanced development programs, $46.6 million; for what we call demonstration validation of the technologies, $83.8 million; for engineering management development, which is actually putting the technologies into the widgets and doing the final operational and developmental testing, $100.8 million; and for overall management of the program, publication of doctrine, training requirements and the training base for chem-bio defense, about $23.9 million, for a total of $361.9 million for research and development. But probably more importantly, we are going to be spending $473.9 million to physically procure new equipment and putting it into the hands of the warfighters in all of those areas I discussed–detection, identification, early warning."

[2] "Congressman, Gosar, thank you. The total energy- related revenues to the Nation are nearly 100 percent. They are–well over 90 percent of the general revenue funds come from royalties, taxes, right-of-way fees, projects related to that. And Navajo Oil and Gas themselves contribute to 10 to 15 percent or more of that total revenue. The other comes from other energy companies,

Supplemental Material (not copyedited or formatted) for: Pamela Ban, Ju Yeon Park, Hye Young You. 2026. "Bureaucrats in Congress: The Politics of Inter-branch Information Sharing."
The Journal of Politics 88(2). DOI: https://doi.org/10.1086/735440.

and our rate is rapidly increasing. I may also comment that relative to the energy delays, our very first Navajo Nation issues–what are called operating agreements, not standard BIA leases–the first operating agreement that the Council approved took over 400 days for BIA approval. The more recent one was still approximately nine months. These type of days, when the company paid out in excess of $4 million to the Nation's general fund for the rights to explore this land, are just economic–huge economic hurdles that we have to overcome."

[3] "This fiscal year we are increasing commodities to the Colombian police–aircraft parts, tools, avionics, field investigative equipment–from $7.4 million to $12.6 million. Training is at $1.5 million. Aircraft operations and so on are doubling from $4.1 million to $8 million. Military assistance would involve $2.5 million in commodities, $1 million in training and $1.5 million in other programs. Judicial sector reform, we are now picking up support for this very important program of $250,000, and we're providing aviation services. We will be providing aviation programs at $14 million, and in addition, new equipment this year involving UH-1H helicopters valued at $10.8 million, Bell 212 helicopters valued at $9 million, and OV-10 Bronco aircraft valued at $84 million. So actually that is a total of $147.8 million."

[4] "Chairman Walberg, first of all, the intent of this regulation is to extend the most basic economic protections to this workforce–the minimum wage and overtime protections. Contrary to your opening statement, the department estimates that the average analyzed costs to employers to familiarize themselves with the regulation would total about $4.7 million over 10 years; and that the increase or transfer of–of transfers to home–of wages to home health care workers in the form of increased minimum wage protections would be approximately $16.1 million; the payment for time spent traveling between patients, approximately $34.7 million; and the payment of overtime premium for hours worked over 40 hour–40 hours in a work week would range between $0 and $180 million per year, on average. So consequently, the impact of this regulation is not $2.8 billion; it is actually rather modest–a modest proposal to extend significant economic protections to this workforce."

[5] "Yes, sir. So, you know, the Corps receives appropriations in different accounts: investigations, construction, and operations and maintenance. And so the numbers that you heard today are only one–they only reflect the Operations and Maintenance account. They don't reflect the Construction and the Investigations account. When you look at all appropriations across all the business lines in 2011, we had: $72.8 million allocated and spent for flood risk management; $15 million for navigation; $61.4 million for hydropower; $13.3 million for environmental stewardship; $800,000 for water supply; $21.6 million for recreation; and $87 million for environmental restoration. So that was last year's budgeted and spent amount, sir."

[6] "That would be terrific. That would be great. The last program I would like to mention real quickly is the State Drinking Water Security Responsibility. Since the events of 2001 as well as the more recent events, hurricanes, wildfires and floods, states have taken on exceptional measures to meet the security and emergency response-related needs of the drinking water community. They provided assistance, training, information and financial support to their water systems and continually work toward integrating security considerations into all aspects of their programs. The appropriated level in fiscal year 2009 was about $5 million or a little less than $100,000 per state,

Supplemental Material (not copyedited or formatted) for: Pamela Ban, Ju Yeon Park, Hye Young You. 2026. "Bureaucrats in Congress: The Politics of Inter-branch Information Sharing."
The Journal of Politics 88(2). DOI: https://doi.org/10.1086/735440.

and states have a tough time understanding why that level has been flat-funded since 2002. And so we respectfully request $7 million in fiscal year 2010 for funding state drinking water security initiatives."

[7] "I'm not sure I have those numbers for seven years. I can tell you that during the last two years, that number is in the range of $147 million of State money. That includes a Clean Water Management Trust Fund. We put about 6.5 percent of remaining funds after the budget is complete into a fund and that's anywhere from $40 to $50, $55 million a year. And, in addition to that, we just, of course, passed the Clean Water Responsibility Act. We've significantly increased our ag share program, working with the farmers on BMPs and so forth; so, $147 million if you total that."

[8] "Another initiative provides funding for ocean conservation. In the refuge program about $400,000 will go to the Palmyra Atoll Research Consortium, and we would also put about $500,000 into the Marine Debris Campaign to help clean up. It is a very serious issue in our coastal refuges. As part of the Department's Safe Borderlands initiative, we have requested $1 million to add six new law enforcement officers in refuges along the southwest border. This would take us from 26 to 32. Now I will turn to discussing our budget request for the Service's programs. For the refuge system, the budget sustains the funding increase of $35.9 million that Congress approved in 2008. And given the difference between the 2008 President's request and the 2009 President's request, I believe that your work last year made a significant impact on OMB to help us sustain that increase."

[9] "The President's budget mark for the CFTC was $130 million. The House Agriculture Subcommittee for Appropriations recently gave us $135 million. As a result of their efforts, we have asked on top of the $130 for an additional $27 million, $21 million to increase our staffing levels by roughly 100 FTEs to get us up to historic levels of where we need to be. Second, the implementation of the farm bill requires us to regulate new markets, known as exempt commercial markets. This Committee helped enact this provision that will require additional staff as well. And, so we have asked for an additional $6 million on top of the $21 million for a total of $27 million."

[10] "Yes. I have the notes of who the entities are. You have got–Health and Human Services was $811 million of the amount. Education was $530 million. The USAID was $169 million. Commerce was $15 million. Energy, $13 million. Labor, $9 million. NASA, $7 million. Then a bunch of other ones were the rest. Keep in mind, our analysis excluded things like Medicaid. It was only limited to certain grant systems, and we looked at the payment systems that were–these were payments made, so $1.6 billion of payments made related to grant programs at those specific agencies."

The least analytical statements (in ascending order)

[1] "I guess I don't know what to think of it. I was surprised by it. I believe that they are friends and–but I don't know."

[2] "Well, there shouldn't be any more. There shouldn't be any more."

[3] "Some of it was, some of it was not. Most of it was."

Supplemental Material (not copyedited or formatted) for: Pamela Ban, Ju Yeon Park, Hye Young You. 2026. "Bureaucrats in Congress: The Politics of Inter-branch Information Sharing."
The Journal of Politics 88(2). DOI: https://doi.org/10.1086/735440.

[4] "Well, there are some that are. There are some that are not."

[5] "Of which they do very, very well. They do it very, very well."

[6] "I do, but I don't have that with me. But we do."

[7] "No, no, no. I won't do that. No. That is for you all."

[8] "But you have to do it, and we are doing it."

[9] "I do. I don't have it with me, but I do."

[10] "We did not have that here. We did not have that here."

These examples are consistent with the features characterizing the most and least analytical paragraphs that were labeled as shown above. This suggests that our machine learning models successfully predicted the analytical scores validating our prediction process.

The statements scoring high tend to contain falsifiable statements frequently involving statistical information and explanations of where revenue and funding for the issue at hand originates.[29]

One additional aspect found here is that highly analytical statements tend to be longer than non-analytical statements, which is not the case for labeled paragraphs because their length was controlled to facilitate human coding. The relationship between the length of statements and their scores is intuitive because the statements that are too short to convey any meaningful information are likely to be evaluated as less analytical.

Finally, we take a further step to validate our final, predicted measurement for analytical information. As we did for the human-coded set, we manually constructed a 5-point scale measurement on a random subset of 120 statements in the corpus that were not labeled by assigning the highest score to the most analytical statements. The correlation coefficient between our manual coding and the analytical scores is 0.876 suggesting that the analytical score captures the concept that we intended to measure very well.

[29]One concern may be that words relating to money or financial matters may bias the measurement of the analytical score. While words relating to money or financial matters do correspond with higher analytical scores, this substantively reflects the type of information that we want to measure for the concept of analytical information—executive agencies are funded by Congress and a main source of discussion and technical information provision is on *how* agencies' budgets are used and the financial cost and impact of their programs. As such, we do not view the correspondence between money- or financial-related words and high analytical information as a problem.